National University of Computer and Emerging Sciences

Lahore Campus

ıalogue & Digital ɔmmunication (EE3010/3003)

Sessional-I Exam

te: September 23, 2026

urse Instructor(s)

Dr. S. M. Sajid (Course Moderator)

Mohsin Yousuf

Total Time (Hrs):	1
Total Marks:	60
Total Questions:	3

oll No _____ Section _____ Student Signature _____

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1. **Attempt all questions and remember to solve parts of the same question together.**
2. **Show all the steps with the help of diagrams and equations.**

LO # 01: ***Explain** the fundamental models of communication systems, signal transmission characteristics, distortions, and energy/power spectral densities. [C2]*

Q1: **[20 marks]**

(a) In a communication channel, **distinguish clearly** between *additive thermal noise* and *signal distortion*. Which of these can be completely corrected using an ideal equalizer? **[6]**

(b) **Classify** the exact mathematical conditions on amplitude response $|H(\omega)|$ and phase response $\theta(\omega)$ for a system to achieve *distortionless transmission*. **[4]**

(c) A deterministic voltage signal given by $g(t) = e^{-3t}u(t)$ is applied to the input of an *ideal low-pass filter (LPF)* with a transfer function: **[10]**

$$H(\omega) = \begin{cases} 2, & |\omega| \le \omega_c \\ 0, & |\omega| > \omega_c \end{cases}$$

1. **Find** the Energy Spectral Density (ESD) $\Psi_g(\omega)$ of the input signal.
2. **Determine** the total energy E_g of the input signal.
3. **Compute** the required cutoff frequency ω_c (in rad/s) such that the output signal contains exactly 95% of the input signal's total energy.

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CLO # 01: Explain the fundamental models of communication systems, signal transmissio characteristics, distortions, and energy/power spectral densities. [C2]

Q2: A periodic power signal $g(t) = 8\cos(100t) + 4\sin(200t)$ passes through a transmission medium modeled as a *first-order RC high-pass* network with a transfer function: **[20 mark**

$$H(\omega) = \frac{j\frac{\omega}{100}}{1 + j\frac{\omega}{100}}$$

(a) Compute the Power Spectral Density (PSD) $S_g(\omega)$ of the input signal using Dirac delta functions. **[8]**

Hint: Exponential Fourier series can be represented as:

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}, \text{where } D_n = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-jn\omega_0 t}\, dt$$

Alternatively, for sums of sinusoids, you may substitute Euler's identities to represent the periodic signal $g(t)$ into its exponential Fourier series:

$$\cos(\omega t) = \frac{e^{j\omega t} + e^{-j\omega t}}{2}, \qquad \sin(\omega t) = \frac{e^{j\omega t} - e^{-j\omega t}}{2j}$$

For computing PSD, you can use: $S_g(\omega) = 2\pi \sum_{n=-\infty}^{\infty} |D_n|^2 \delta(\omega - \omega_0)$

(b) Determine the total average power P_i of the input signal. **[4]**

(c) Find the total average power P_o of the output signal after filtering. **[8]**

Hint: Power transfer function relationship between input and output is:

$$S_o(\omega) = |H(\omega)|^2 S_g(\omega)$$

CLO # 02: Calculate parameters like bandwidth, power efficiency, transmission efficiency, and modulation index for linear (AM, SSB, VSB, QAM) and non-linear (FM, PM) analog modulation systems.[C3]

Q3: A hardware designer implements a switching amplitude modulator using a balanced bridge configuration containing non-linear (NL) matching elements. The operational behavior of the system is simplified and modeled via the functional block diagram layout provided in Figure 1. **[20 marks]**

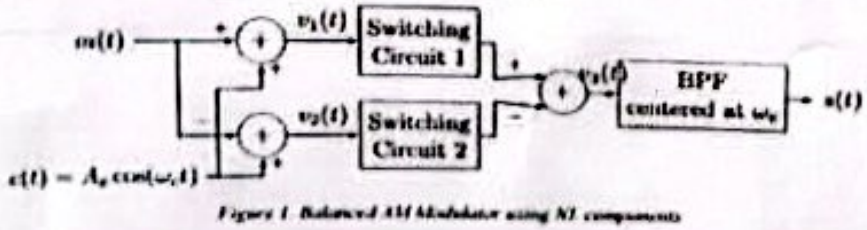


Figure 1 Balanced AM Modulator using NL components

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The high-power carrier signal $c(t) = A_c \cos(\omega_c t)$ controls the bridge networks, effectively forcing the operational path behavior to act as an ideal square-wave switching function $w(t)$ flipping between 1 and 0 at carrier frequency intervals. The square wave has a Fourier series representation of:

$$w(t) = \frac{1}{2} + \frac{2}{\pi} \sum_{n=1,3,5,\ldots}^{\infty} \frac{(-1)^{\frac{n-1}{2}}}{n} \cos(n\omega_c t)$$

The output signals from the upper and lower switching bridge branches are defined precisely by $v_{o1}(t) = v_1(t) \cdot w(t)$ and $v_{o2}(t) = v_2(t) \cdot w(t)$ respectively.

(a) **Determine** the mathematical expressions for the pre-switching voltages $v_1(t)$ and $v_2(t)$ entering the bridge branches. [4]

(b) **Apply** the square-wave switching modulation model to derive the expanded structural time-domain expression for the post-subtraction signal $v_3(t)$. [10]

(c) Track the spectral components through the ideal Bandpass Filter (BPF) to prove that the final output s(t) constitutes a standard Double-Sideband Suppressed Carrier (DSB-SC) AM wave. **Determine** its effective scaling amplitude. [6]

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